

# Introduction to non-linear filtering

# Sharpening and Enhancement



Original



Processed

**PDE and variational approaches for stable sharpening and denoising.**

<http://guygilboa.eew.technion.ac.il/researches/sharpening-and-enhancement/>

# ANISOTROPIC DIFFUSION USING POWER WATERSHEDS

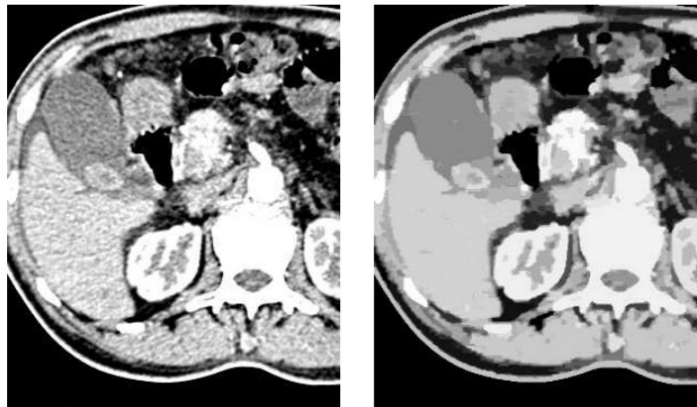
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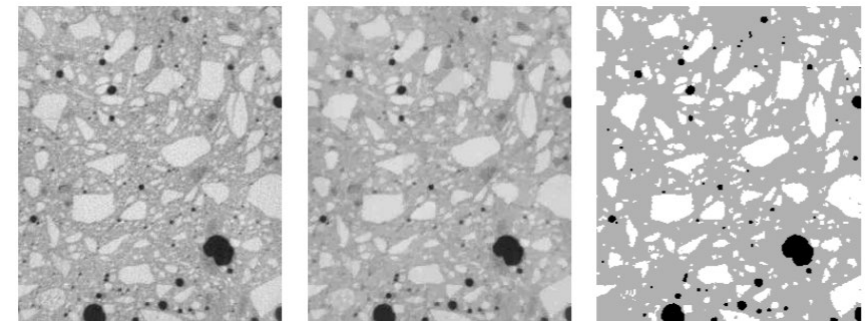
leo.grady@siemens.com



(a) Original image

(b) PW result

**Fig. 3.** Filtering of a liver image by power watershed. Here noise and small vessels are both removed, leading to a result which may be used as a first step before segmentation.



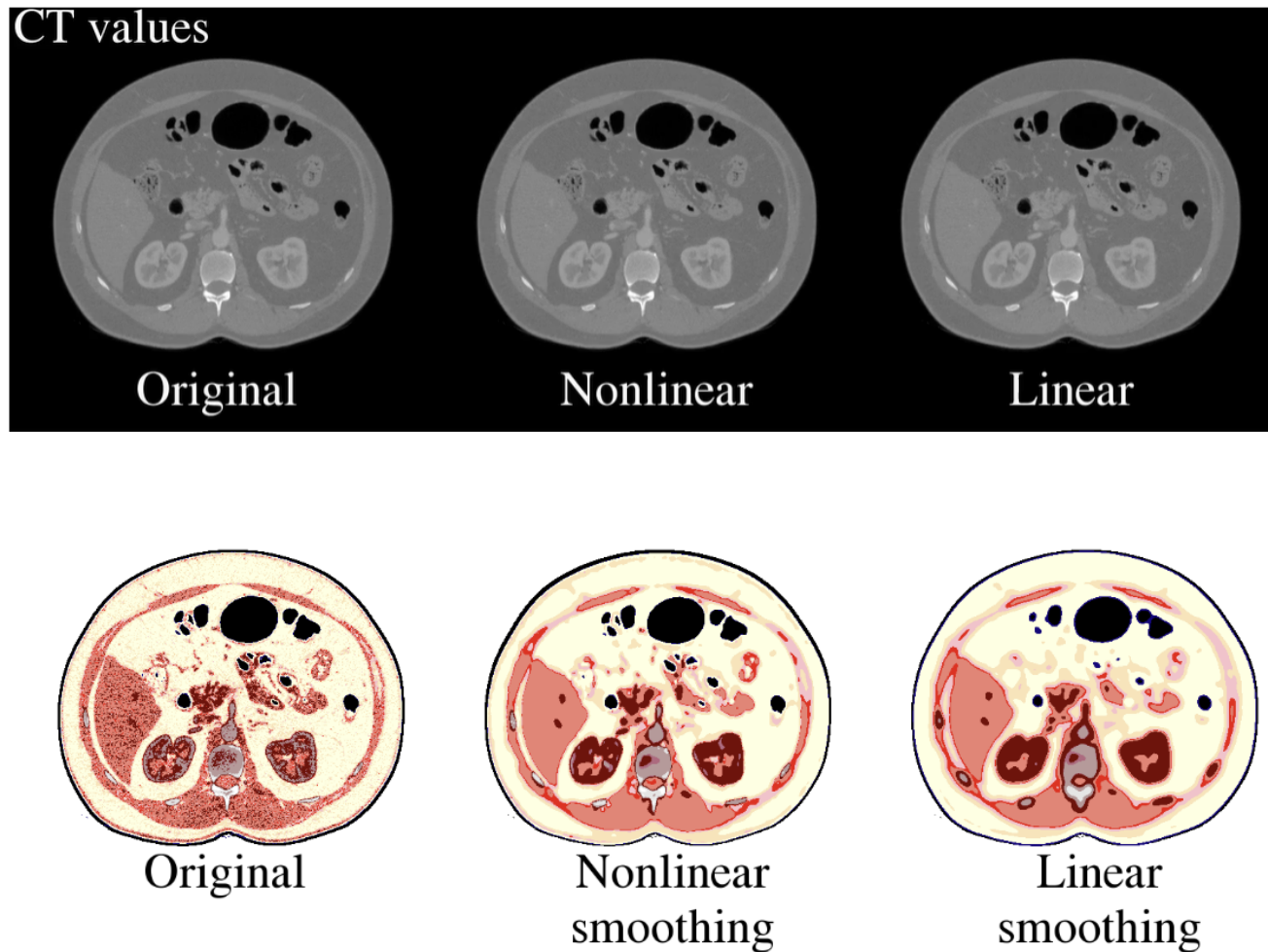
(a) Original image

(b) PW result (6 iter.)

(c) Segmentation

**Fig. 4.** The segmentation of concrete images in three classes (bubbles in black, stones in light grey) is useful for the study of the material's mechanical properties. The segmentation is obtained by two thresholds of the filtered image by PW.

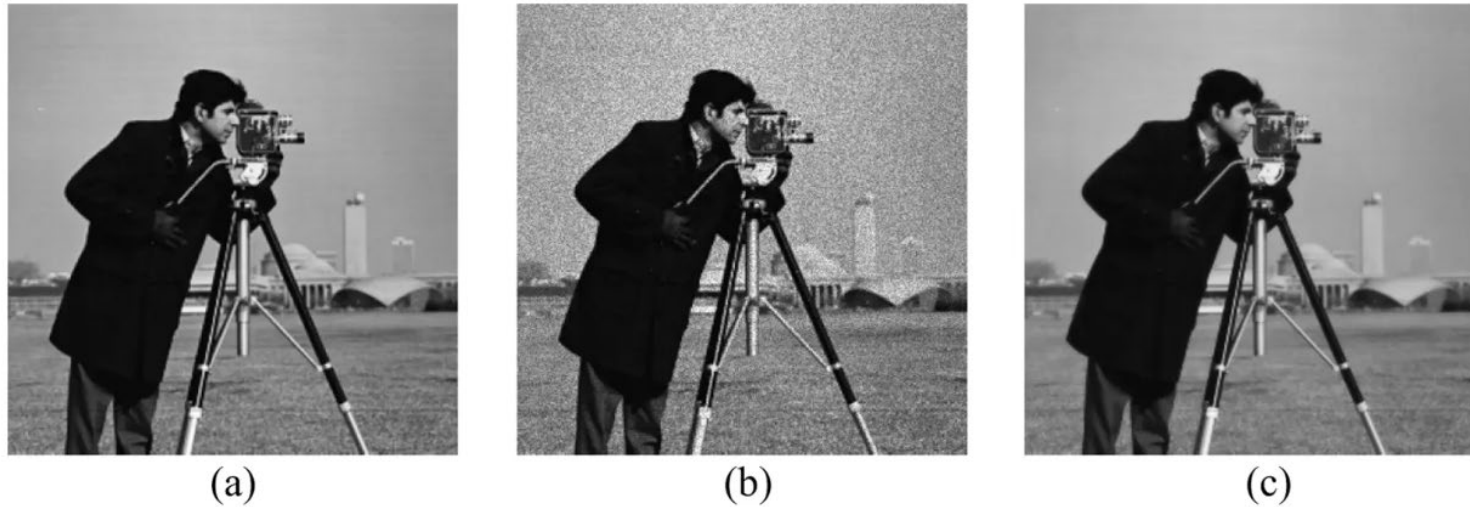
# Image Smoothing by Nonlinear Diffusion



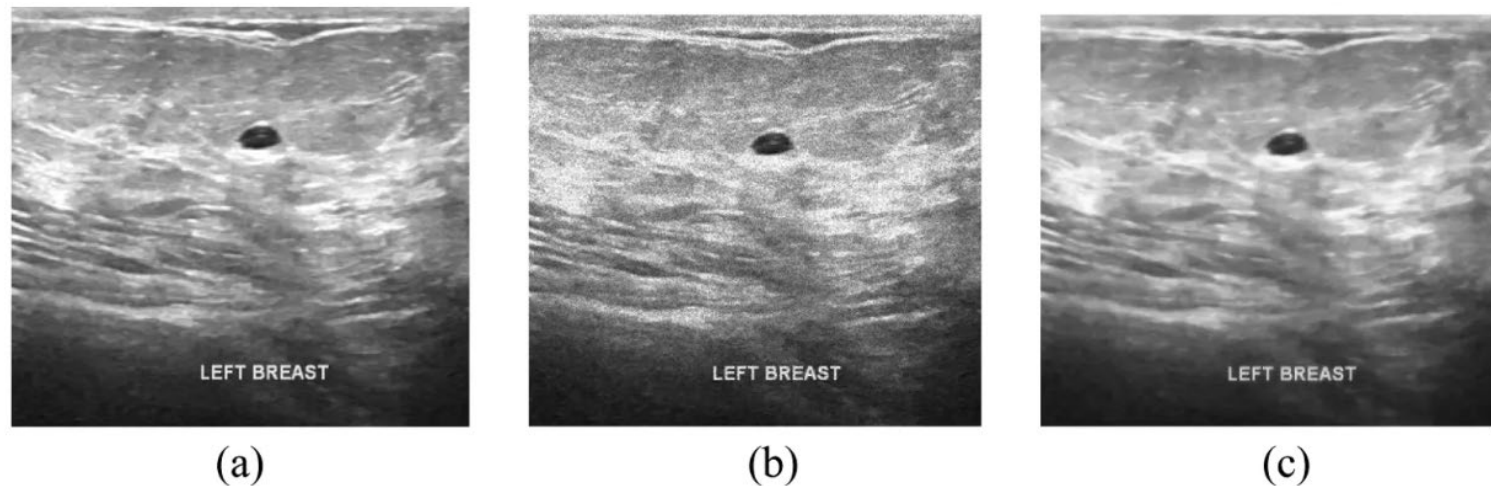
**Figure 1:** When comparing a linear image smoothing to a nonlinear one we can see that it can reduce noise while preserving edges and important structures. This example is of a scalar valued CT image that has been colored for illustration purposes.



# Extended Speckle Reduction Anisotropic Diffusion Filter

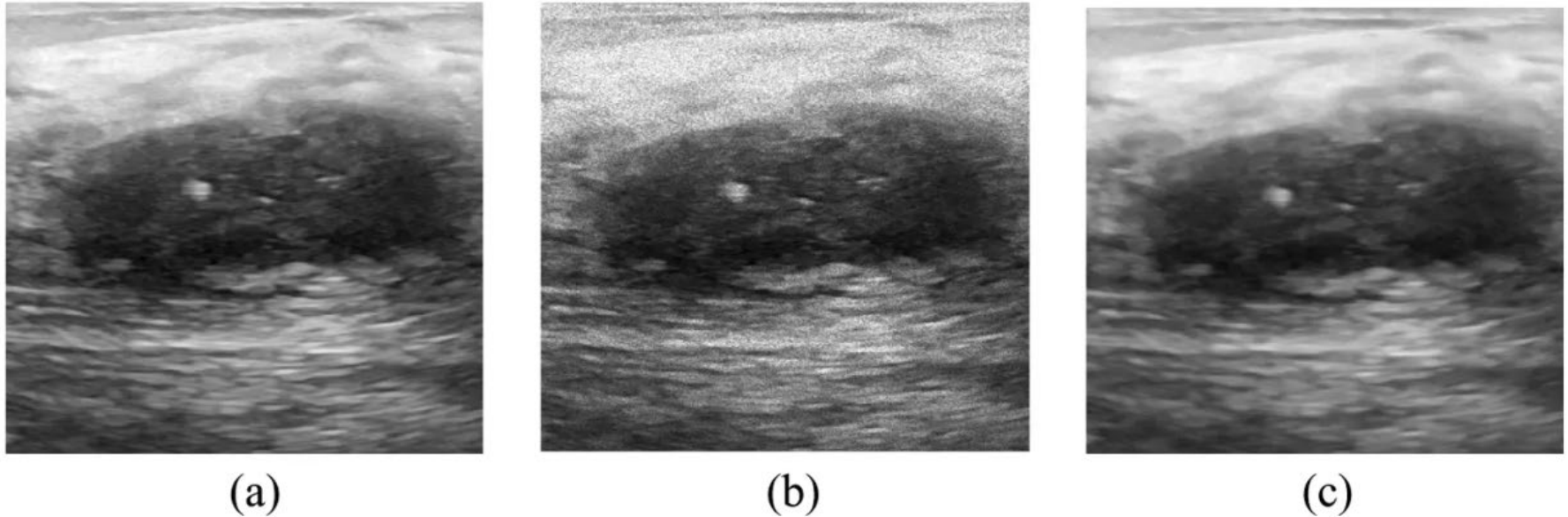


**Figure 4:** Cameraman Image: (a) Original (b) With noise variance 0.35 (c) Denoised output using extended SRAD filter



**Figure 6:** Breast Ultrasound image (Benign): (a) Original (b) With noise variance 0.3 (c) Denoised output using extended SRAD filter

# Extended Speckle Reduction Anisotropic Diffusion Filter



**Figure 7:** Breast Ultrasound image (Malignant): (a) Original (b) With noise variance 0.3 (c) Denoised output using extended SRAD filter

# Rudin-Osher-Fatemi Model Captures Infinity and Beyond

## Rudin–Osher–Fatemi PDE [\[edit\]](#)

Suppose that we are given a noisy image  $f$  and wish to compute a denoised image  $u$  over a 2D space. ROF showed that the minimization problem we are looking to solve is:<sup>[5]</sup>

$$\min_{u \in \text{BV}(\Omega)} \|u\|_{\text{TV}(\Omega)} + \frac{\lambda}{2} \int_{\Omega} (f - u)^2 dx$$

where  $\text{BV}(\Omega)$  is the set of functions with [bounded variation](#) over the domain  $\Omega$ ,  $\text{TV}(\Omega)$  is the total variation over the domain, and  $\lambda$  is a penalty term. When  $u$  is smooth, the total variation is equivalent to the integral of the gradient magnitude:

$$\|u\|_{\text{TV}(\Omega)} = \int_{\Omega} \|\nabla u\| dx$$

where  $\|\cdot\|$  is the [Euclidean norm](#). Then the objective function of the minimization problem becomes:

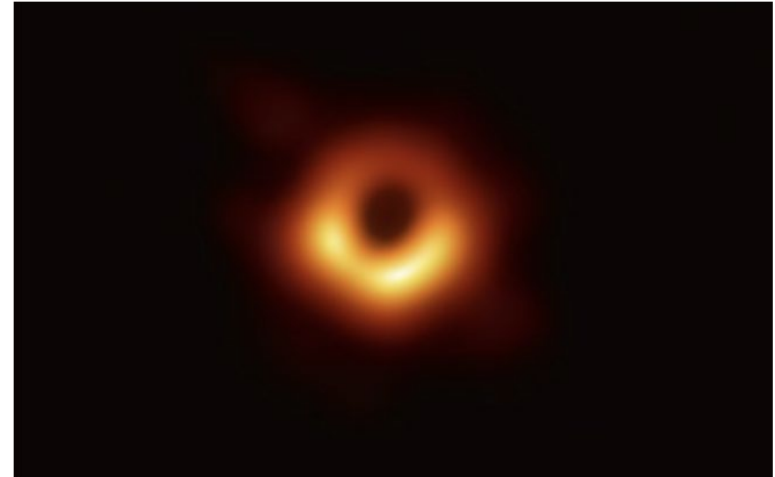
$$\min_{u \in \text{BV}(\Omega)} \int_{\Omega} \left[ \|\nabla u\| + \frac{\lambda}{2} (f - u)^2 \right] dx$$

From this functional, the Euler-Lagrange equation for minimization – assuming no time-dependence – gives us the nonlinear [elliptic partial differential equation](#):

$$\begin{cases} \nabla \cdot \left( \frac{\nabla u}{\|\nabla u\|} \right) + \lambda(f - u) = 0, & u \in \Omega \\ \frac{\partial u}{\partial n} = 0, & u \in \partial\Omega \end{cases}$$

For some numerical algorithms, it is preferable to instead solve the time-dependent version of the ROF equation:

$$\frac{\partial u}{\partial t} = \nabla \cdot \left( \frac{\nabla u}{\|\nabla u\|} \right) + \lambda(f - u)$$



Credit: Event Horizon Telescope collaboration et al.

Source: <https://www.ipam.ucla.edu/news/rudin-osher-fatemi-model-captures-infinity-and-beyond/>  
[https://en.wikipedia.org/wiki/Total\\_variation\\_denoising](https://en.wikipedia.org/wiki/Total_variation_denoising)

(not examined)

# Introduction to non-linear filtering

## References:

- (1) P. Perona and J. Malik, *Scale-space and edge detection using anisotropic diffusion*, IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 12, No. 7, pp. 629-639, 1990.
- (2) Joachim Weickert, *A review of nonlinear diffusion filtering*, Scale-Space Theory in Computer Vision, Lecture Notes in Computer Science, Vol. 1252, Springer, Berlin, pp. 3-28, 1997.
- (3) G. Gerig, *Nonlinear Anisotropic Filtering of MRI Data*, IEEE Transactions on Medical Imaging, Vol. 11, No. 2, pp. 221-232, 1992.



# Roadmap

1. Filtering based on Gaussian low-pass filter
2. Concepts of diffusion
3. Linear diffusion filtering
4. Non-linear diffusion filtering

# Filtering based on Gaussian low-pass filter

## 1. Low-pass Gaussian filtering of a 2D image

$$u(\vec{x}, t) = \begin{cases} I(\vec{x}) & (t = 0) \\ (G_{\sqrt{2t}} * I)(\vec{x}) & (t > 0) \end{cases}$$

where  $\vec{x} = (x, y) =$  Position vector

$I(\vec{x}) =$  Initial image

$u(\vec{x}, t) =$  Filtered image

$*$  = Convolution operator

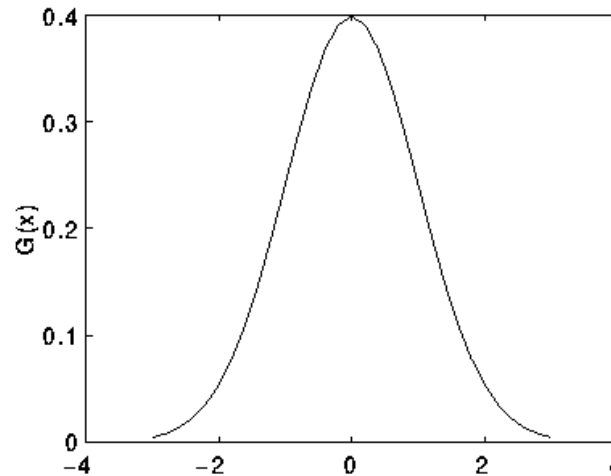
$$G_{\sigma}(\vec{x}) = \frac{1}{2\pi\sigma^2} \cdot \exp\left(\frac{-|\vec{x}|^2}{2\sigma^2}\right)$$

# Filtering based on Gaussian low-pass filter

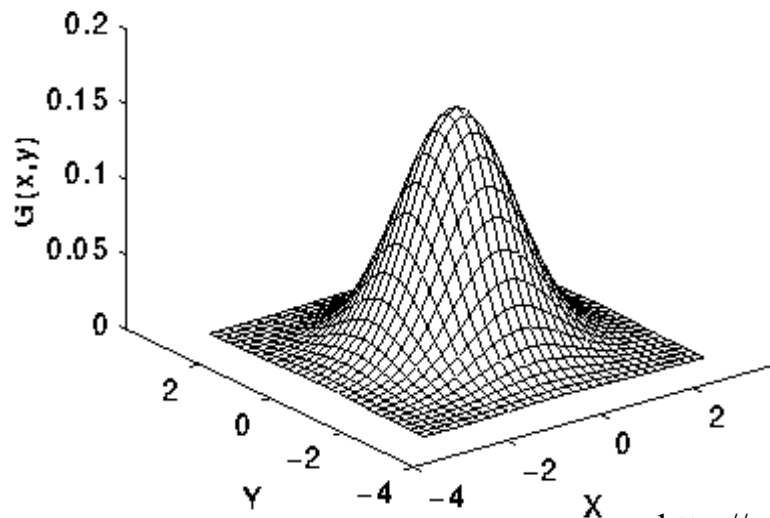
2. Gaussian filtering – it filters or can smooth an initial image  $I$  by filtering (convolving) the image with a Gaussian filter.
3. Filtering (convolution) represents the weighted sum of local intensity values, in which the weights are determined by a Gaussian distribution with higher values in the center and lower values in the tails.

# Filtering based on Gaussian low-pass filter

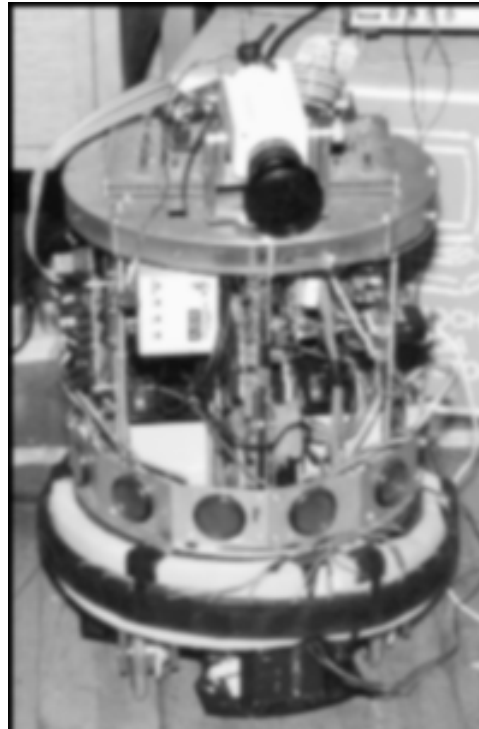
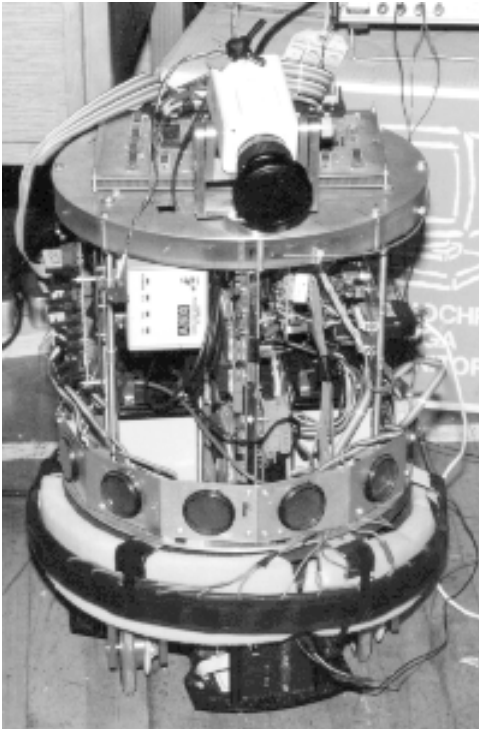
4. 1D Gaussian distribution: zero mean and SD = 1



5. 2D Gaussian distribution: zero mean and SD = 1



# Filtering based on Gaussian low-pass filter



$\sigma = 1$  (5X5 kernel size)       $\sigma = 2$  (9X9 kernel size)

Gaussian smoothing: <http://www.dai.ed.ac.uk/HIPR2/gsmooth.htm>



# Filtering based on Gaussian low-pass filter

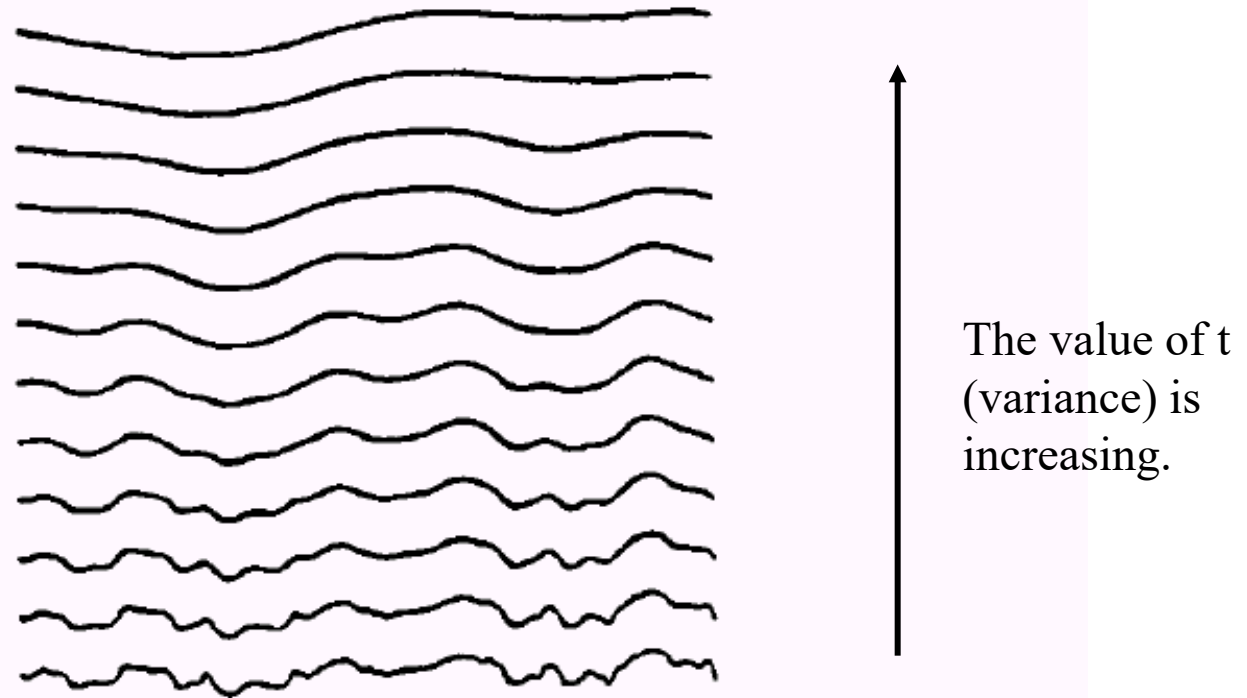


Fig. 1. A family of 1-D signals  $I(x, t)$  obtained by convolving the original one (bottom) with Gaussian kernels whose variance increases from bottom to top (adapted from Witkin [21]).

A family of smoother image intensity profiles is formed by filtering (convolving) the original image with a Gaussian filter of increasing variance.

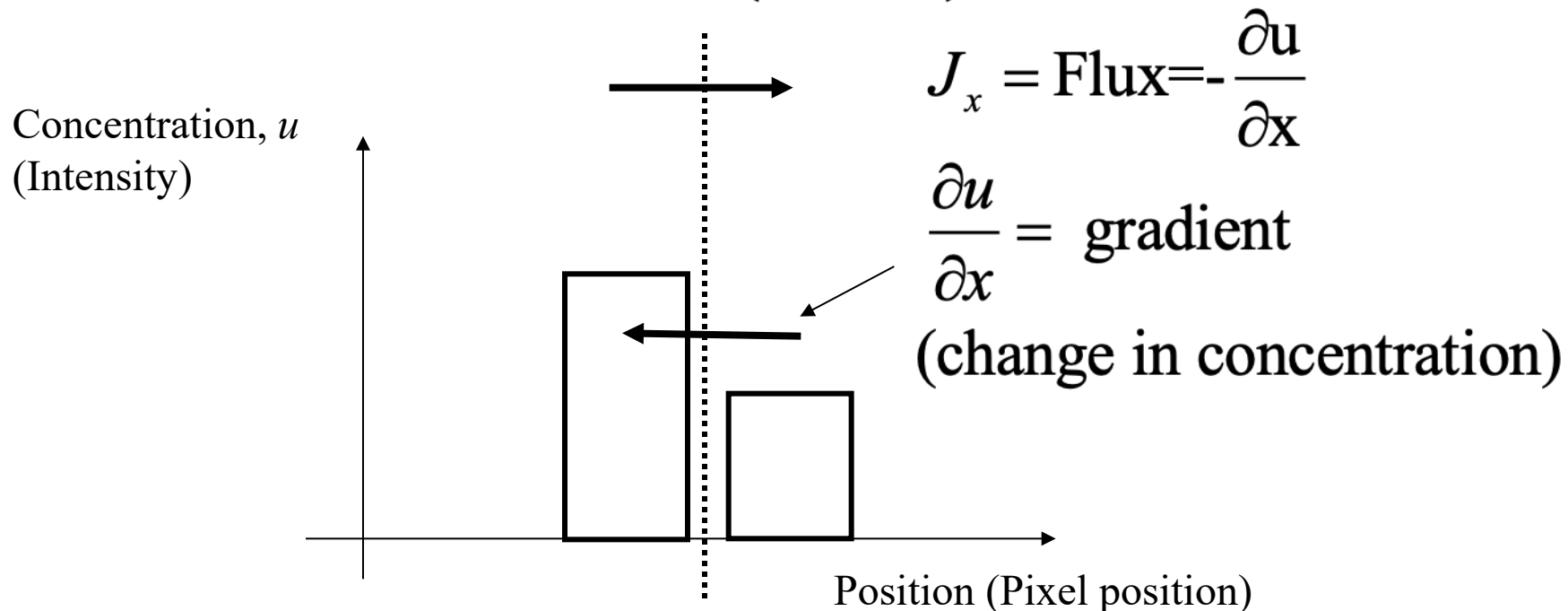
# Concepts of diffusion

1. The first law of diffusion – mass flux (diffusion) is proportional to the concentration gradient (change in concentration)

$$\vec{J} = -D \nabla u$$

$$(J_x, J_y)^T = -D \left( \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right)^T$$

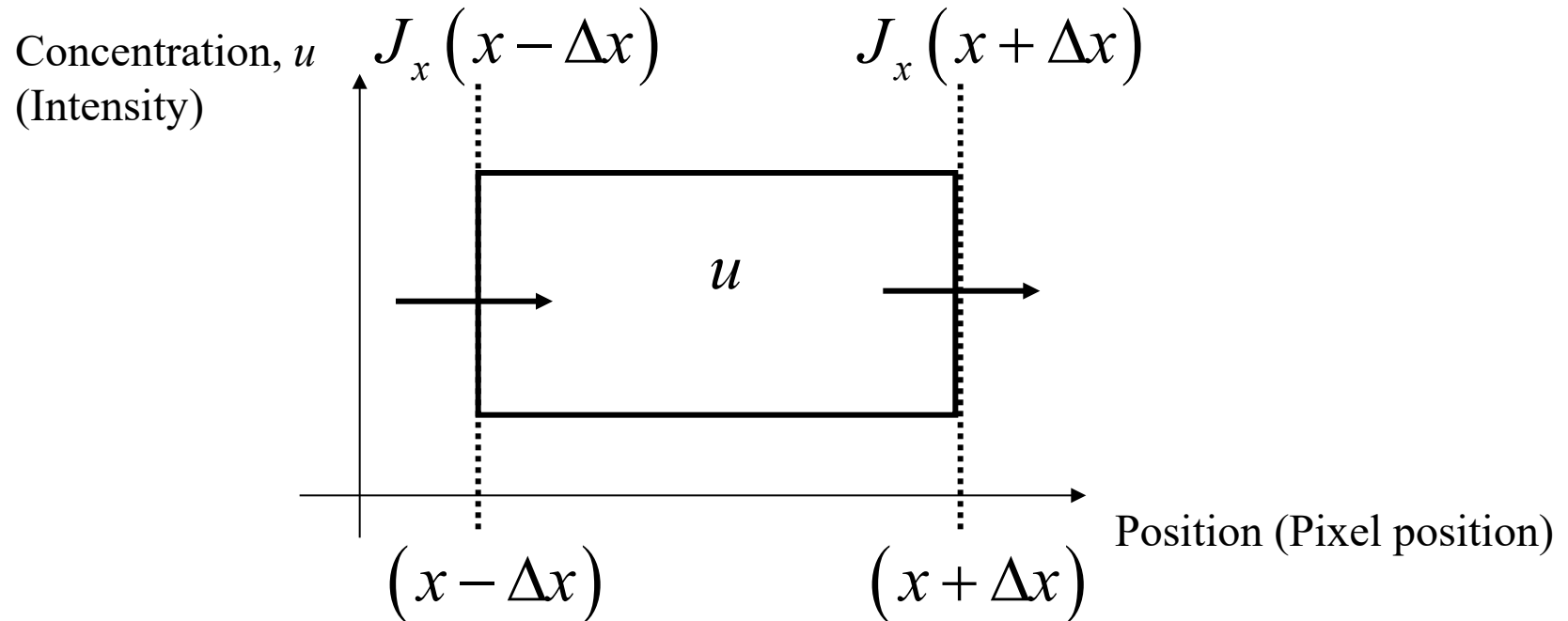
$D$  = diffusion tensor  
It controls the rate and directionality of diffusion



# Concepts of diffusion

2. The second law of diffusion – the rate of accumulation of concentration within a volume is proportional to the change of local concentration gradient (continuity equation).

For example, for 1D 
$$\frac{\partial u}{\partial t} = -\frac{\partial J_x}{\partial x}$$



# Concepts of diffusion

## 3. Equation of diffusion (Heat equation)

E.g., for 1D  $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$  where  $D =$  diffusivity  
or Conductance  
or Diffusion constant

E.g., for 2D  $\frac{\partial u}{\partial t} = \text{div}(D \nabla u)$

where  $D =$  diffusivity tensor

$$\text{div}(\vec{x}) = \nabla \cdot \vec{x}$$

# Concepts of diffusion

4. Linear diffusion filtering:  $D = I$  (Identity matrix), it means heat (intensity) diffuses according to

$$\frac{\partial u}{\partial t} = \text{div}(\nabla u) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \quad \text{Initial condition} \quad u(\vec{x}, t = 0) = I(\vec{x})$$

(Remark: The low-pass Gaussian filtering can be viewed as the solution of the heat equation.)

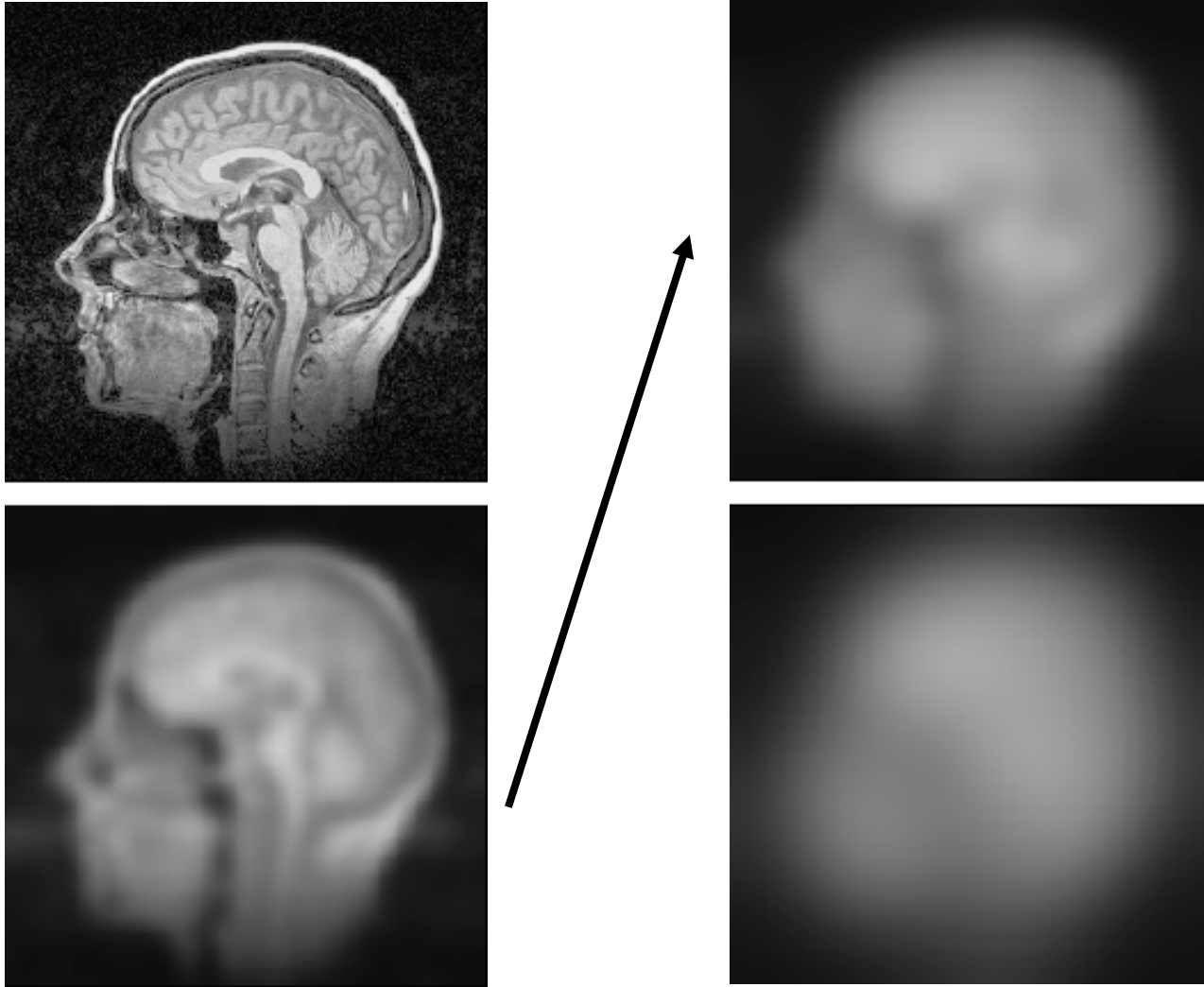
# Linear diffusion filtering

1. Assume that  $D = I$  Identity Matrix
2. Heat equation (2D)

$$\frac{\partial u}{\partial t} = \Delta u$$
$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

Initial condition  $u(\vec{x}, t = 0) = I(\vec{x})$

# Linear diffusion filtering



Linear diffusion examples:  $t=0, 12.5, 50, 200$ .



# Linear diffusion filtering

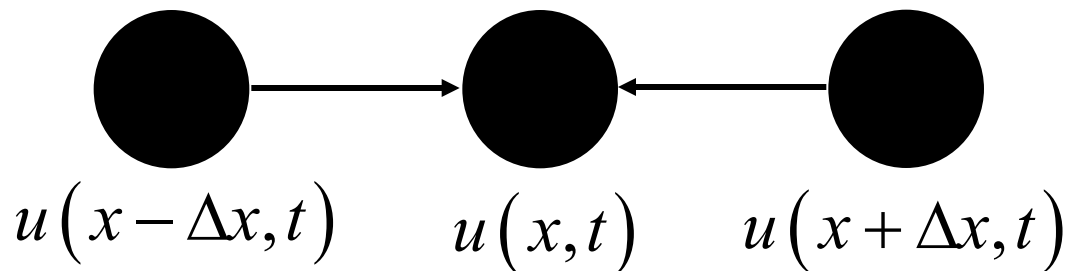
3. Implementation. For example, in 1D

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

Initial condition  $u(x, t = 0) = I(x)$

$$u(x, t + \Delta t) = u(x, t)$$

$$+ \frac{\Delta t}{(\Delta x)^2} \left( u(x + \Delta x, t) - 2u(x, t) + u(x - \Delta x, t) \right)$$



# Linear diffusion filtering

4. Gaussian smoothing / linear diffusion filtering reduces noise, but also blurs important features such as edges and thus makes them harder to identify (see Fig. 1 on next page).
5. Gaussian smoothing / linear diffusion filtering dislocates edges when moving from finer to coarser scales (see Fig. 2 on next page).

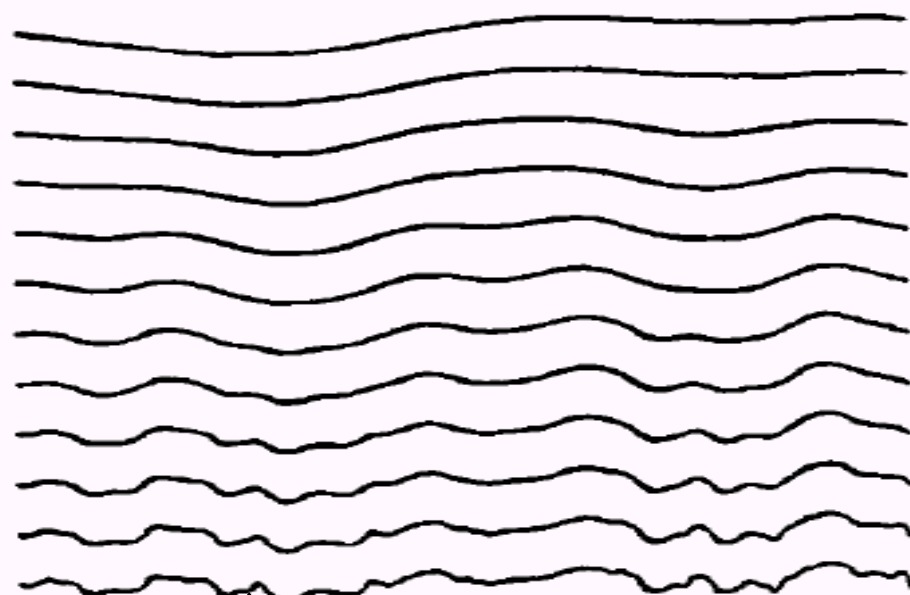


Fig. 1. A family of 1-D signals  $I(x, t)$  obtained by convolving the original one (bottom) with Gaussian kernels whose variance increases from bottom to top (adapted from Witkin [21]).



Fig. 2. Position of the edges (zeros of the Laplacian with respect to  $x$ ) through the linear scale space of Fig. 1 (adapted from Witkin [21]).

# Non-linear diffusion filtering

1. We want to encourage smoothing within a region in preference to smoothing across the boundaries.
2. This could be achieved by setting the diffusivity to be 0 on the boundary and to be non-zero in the interior of the region.

E.g., for 2D 
$$\frac{\partial u}{\partial t} = \operatorname{div}(D \nabla u)$$

where 
$$D = g(|\nabla u|)$$

$$g(|\nabla u|) = \exp\left(-\left(\frac{|\nabla u|}{K}\right)^2\right) \quad \text{or} \quad g(|\nabla u|) = \frac{1}{1 + \left(\frac{|\nabla u|}{K}\right)^2}$$

# Non-linear diffusion filtering

3. Implementation. For example, in 2D

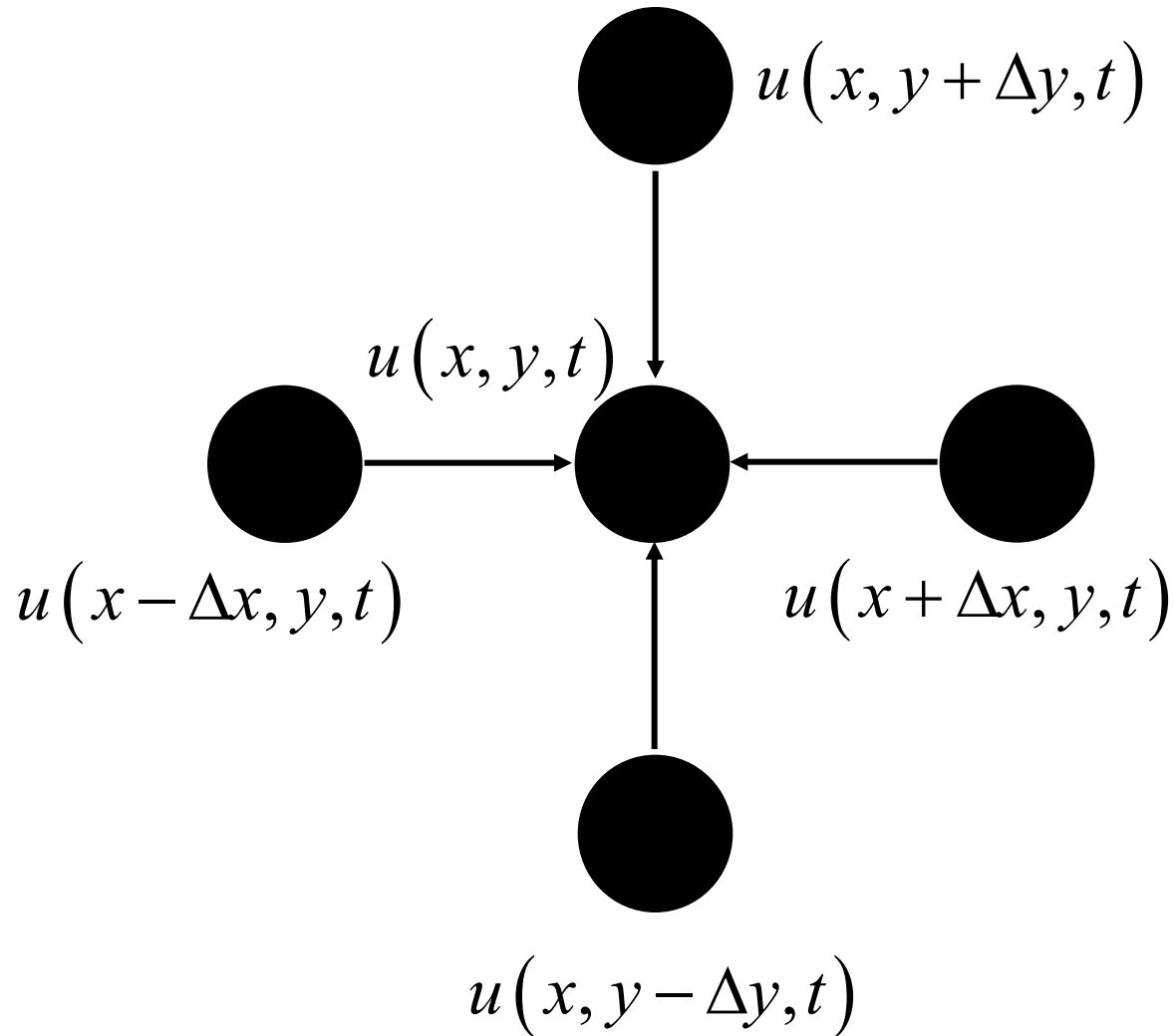
$$\frac{\partial u}{\partial t} = \operatorname{div}(g \nabla u)$$

Initial condition  $u(\vec{x}, t = 0) = I(\vec{x})$

$$u(x, y, t + \Delta t) = u(x, y, t)$$

$$+ \Delta t \left[ (\Phi_N - \Phi_S) + (\Phi_E - \Phi_W) \right]$$

# Non-linear diffusion filtering



# Non-linear diffusion filtering

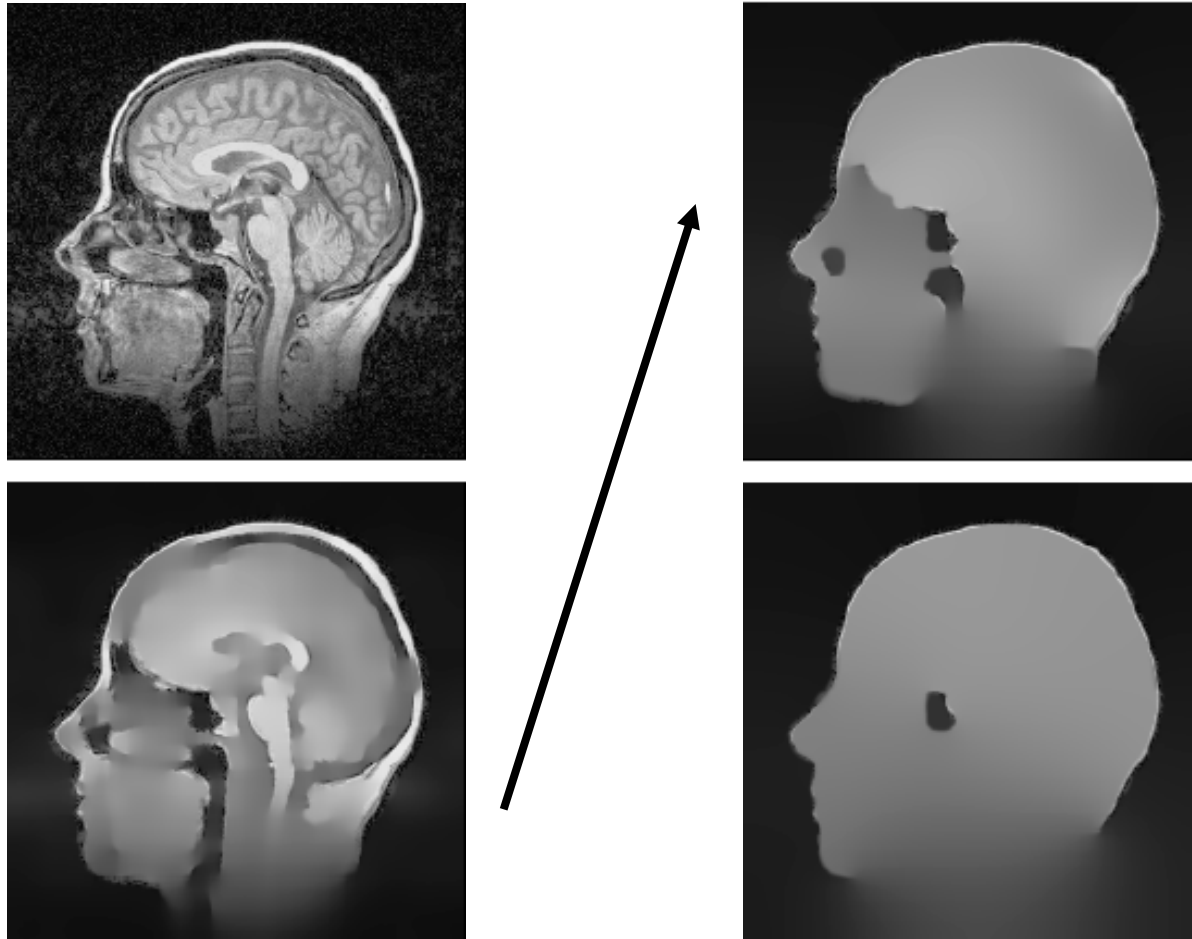
$$\Phi_N = g \left( \left| \nabla u \left( x, y + \frac{\Delta y}{2}, t \right) \right| \right) \left( \frac{u(x, y + \Delta y, t) - u(x, y, t)}{\Delta y^2} \right)$$

$$\Phi_S = g \left( \left| \nabla u \left( x, y - \frac{\Delta y}{2}, t \right) \right| \right) \left( \frac{u(x, y, t) - u(x, y - \Delta y, t)}{\Delta y^2} \right)$$

$$\Phi_E = g \left( \left| \nabla u \left( x + \frac{\Delta x}{2}, y, t \right) \right| \right) \left( \frac{u(x + \Delta x, y, t) - u(x, y, t)}{\Delta x^2} \right)$$

$$\Phi_W = g \left( \left| \nabla u \left( x - \frac{\Delta x}{2}, y, t \right) \right| \right) \left( \frac{u(x, y, t) - u(x - \Delta x, y, t)}{\Delta x^2} \right)$$

# Non-linear diffusion filtering



Non-linear diffusion examples:  $K=3$ ,  $t=0, 40, 400, 1500$ .



Linear diffusion



$$\frac{\partial u}{\partial t} = \Delta u$$

Non-linear diffusion



$$\frac{\partial u}{\partial t} = \operatorname{div}(g \nabla u)$$



(a)

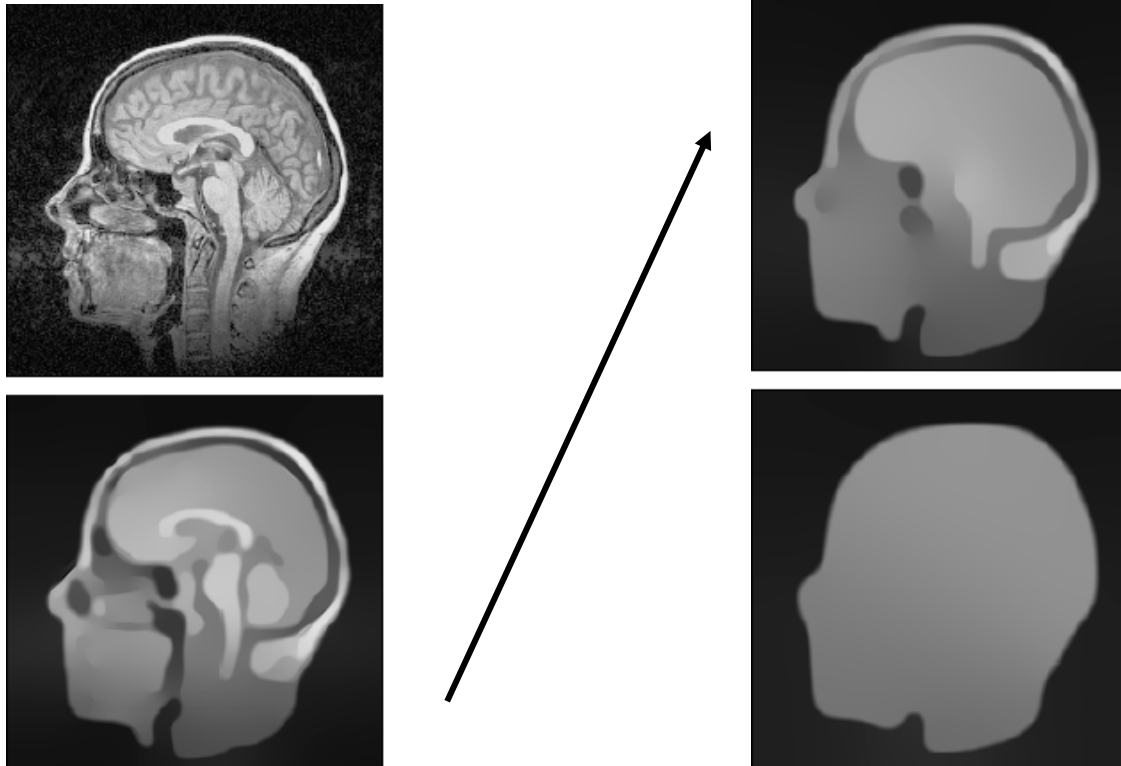


(a)



(b)

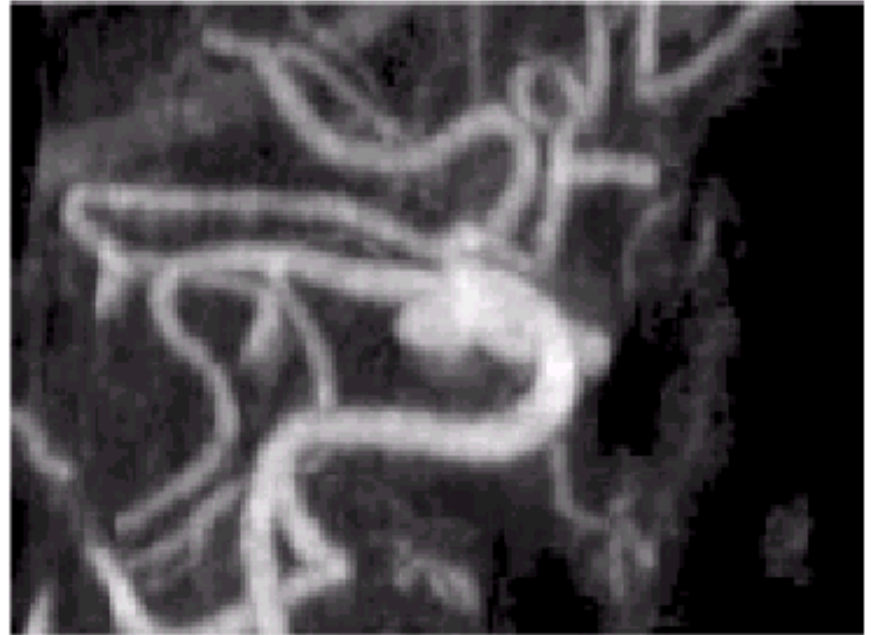
# Edge-enhancing anisotropic diffusion



# Edge-enhancing anisotropic diffusion



Enhancement of tubular structures  
(Karl Krissian, et. al.)



Enhancement of vessels  
(Karl Krissian, et. al.)

# Coherence-enhancing anisotropic diffusion



**Fig. 4.** Coherence-enhancing anisotropic diffusion of a fingerprint image. (a) LEFT: Original image,  $\Omega = (0, 256)^2$ . (b) RIGHT: Filtered,  $\sigma = 0.5$ ,  $\rho = 4$ ,  $t = 20$ . From [74].

# Coherence-enhancing anisotropic diffusion

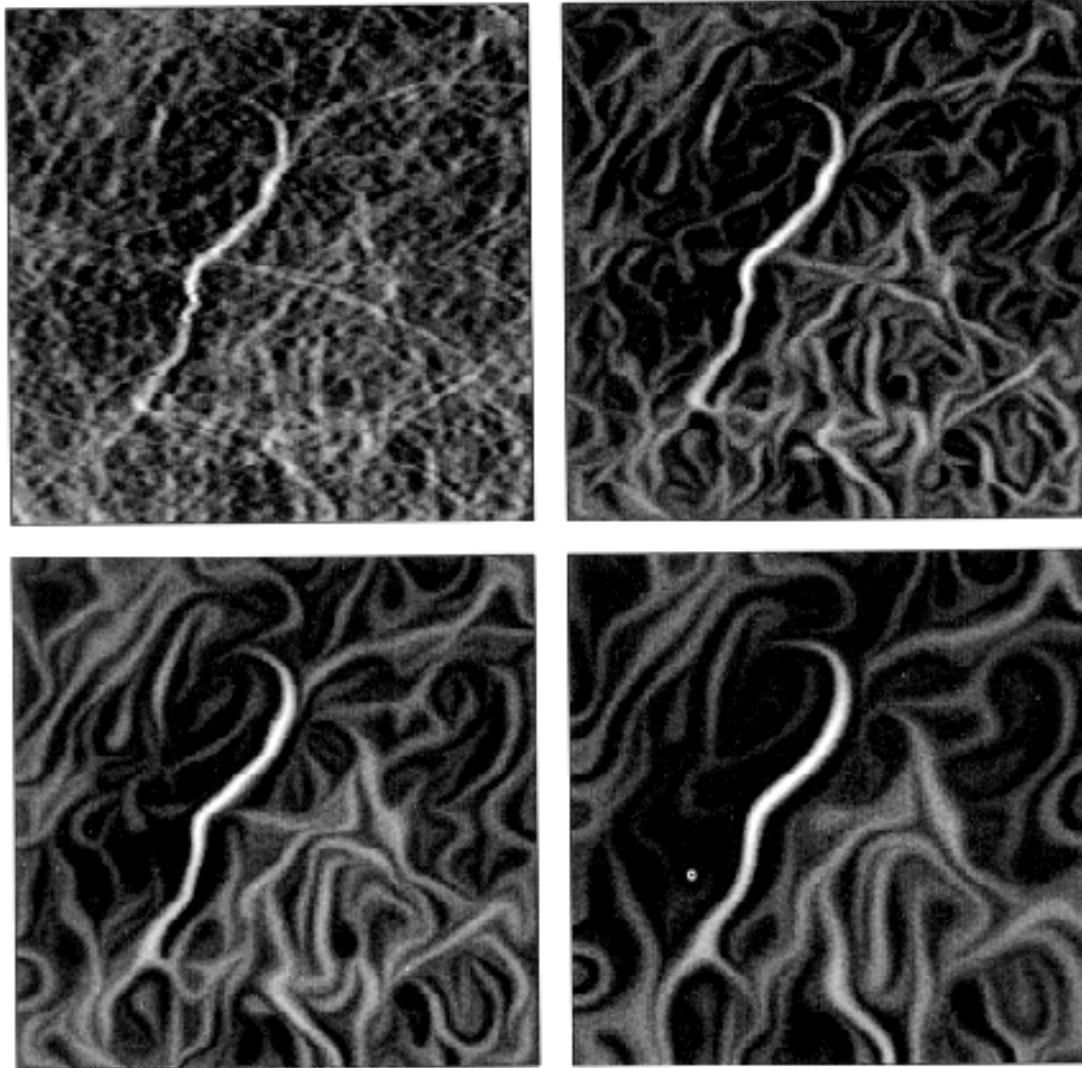


Figure 5.12: Scale-space behaviour of coherence-enhancing diffusion ( $\sigma = 0.5$ ,  $\rho = 2$ ). (a) TOP LEFT: Original fabric image,  $\Omega = (0, 257)^2$ . (b) TOP RIGHT:  $t = 20$ . (c) BOTTOM LEFT:  $t = 120$ . (d) BOTTOM RIGHT:  $t = 640$ .

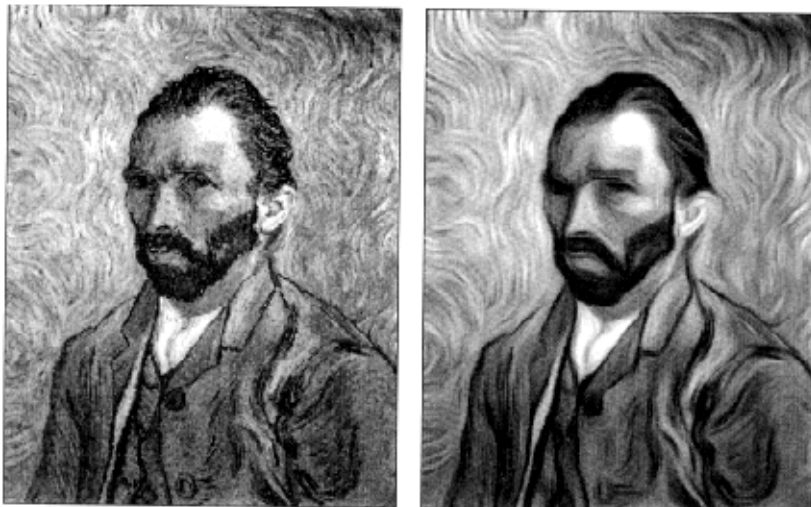


Figure 5.14: Image restoration using coherence-enhancing anisotropic diffusion. (a) LEFT: "Selfportrait" by van Gogh (Saint-Rémy, 1889, Paris, Musée d'Orsay),  $\Omega = (0, 215) \times (0, 275)$ . (b) RIGHT: Filtered,  $\sigma = 0.5$ ,  $\rho = 4$ ,  $t = 6$ .

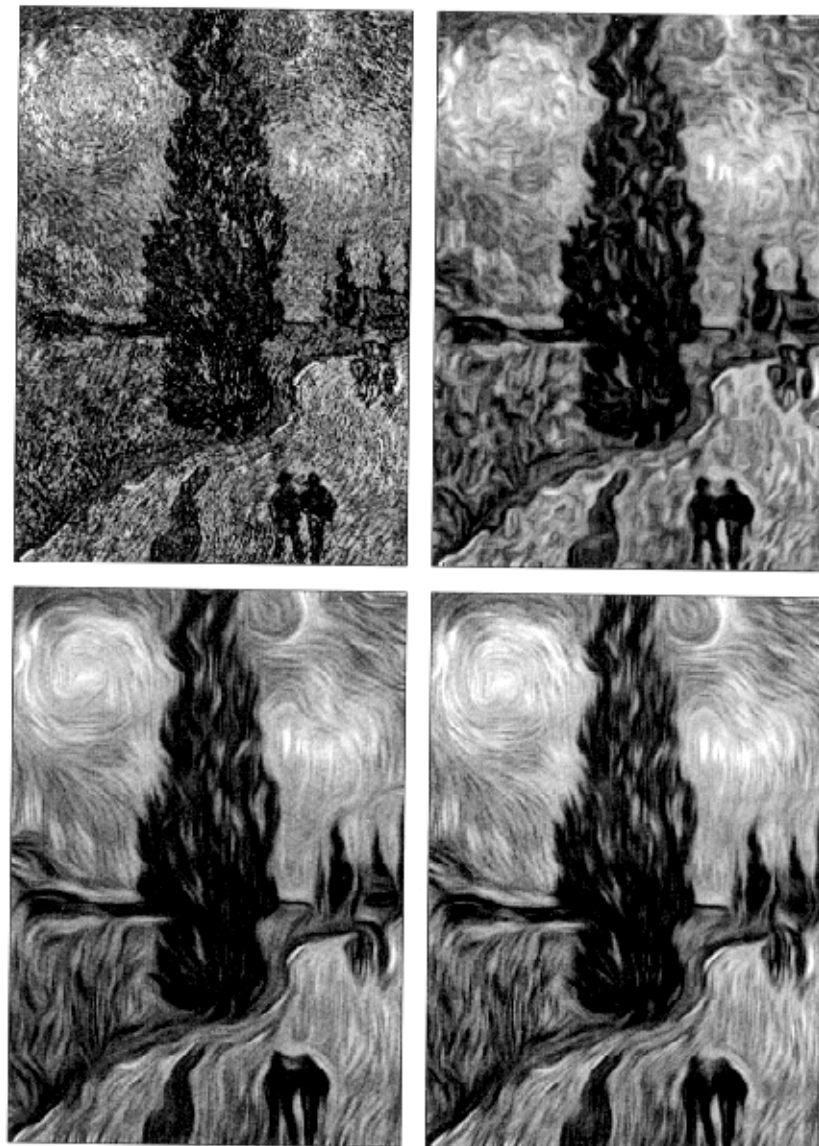


Figure 5.15: Impact of the integration scale on coherence-enhancing anisotropic diffusion ( $\sigma = 0.5$ ,  $t = 8$ ). (a) TOP LEFT: "Road with Cypress and Star" by van Gogh (Auvers-sur-Oise, 1890; Otterlo, Rijksmuseum Kröller-Müller),  $\Omega = (0, 203) \times (0, 290)$ . (b) TOP RIGHT: Filtered with  $\rho = 1$ . (c) BOTTOM LEFT:  $\rho = 4$ . (d) BOTTOM RIGHT:  $\rho = 6$ .